

PATH-INTEGRAL FORMULATION OF CHIRAL INVARIANT FERMION MODELS IN TWO DIMENSIONS

K. FURUYA, R.E. GAMBOA SARAVI^{1,2} and F.A. SCHAPOSNIK¹

Laboratoire de Physique Théorique et Hautes Energies, Paris, France*

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We study the Thirring and chiral-invariant Gross-Neveu (CGN) models using the functional integral method. By introducing an auxiliary vector field we disclose a relation with two-dimensional gauge theories coupled to fermions and then extend a technique based on a chiral change in the functional variables to study purely fermionic models.

We obtain the exact Klaiber solution for the massless Thirring model (for spin $\frac{1}{2}$) in a very simple way and we then extend our technique to investigate the CGN model. We show the factorization of a free fermionic part at the level of Green functions on very general grounds. We then impose certain restrictions on the behavior of the fields – which render our treatment exact only in the zero winding number sector, but allow the computation of the U(1) part of the CGN Green functions exactly, showing, in particular, its complete decoupling from the color part and the almost long-range order behavior in the infrared region.

In our approach, the non-triviality of the jacobian arising from the chiral transformation – directly related to the topological density and the axial anomaly – appears to be crucial for the functional integral treatment of these models.

1. Introduction

Quantum field theoretical models in two space-time dimensions have proven to be a fruitful laboratory to test various phenomena like dynamical mass generation, asymptotic freedom and quark confinement, all features to be welcomed in a realistic theory of strong interactions.

Of particular interest are those models describing self-interacting fermions (like the Thirring [1] and chiral invariant Gross-Neveu models [2]) or fermions coupled to gauge fields (the Schwinger model [3] and two-dimensional quantum chromodynamics [4]) since exact solutions are known in some cases and have given relevant insights into the phenomena mentioned above.

¹ Member of CIC Buenos Aires, Argentina.

² Laboratorio de Física Teórica, Universidad Nacional de La Plata, Argentina.

* Laboratoire Associé au C.N.R.S. Postal Address: Université Pierre et Marie Curie (Paris VI), Tour 16–1er étage-4 place Jussieu, 75230 Paris Cedex 05, France

For the Thirring model and QED_2 with massless fermions (the Schwinger model) an operator construction has lead to exact solutions [5, 6] in the sense that all the correlation functions can be calculated. The same method, applied to the chiral invariant Gross-Neveu model (CGN) yielded the knowledge of the S -matrix [7]. On the other hand, the use of the Bethe ansatz technique has lead to the diagonalization of the hamiltonian and the determination of the physical spectrum and S -matrix for the massive Thirring [8] and CGN [9] models.

Among the problems still unsolved, there is the extension of the operator techniques, successfully applied to QED_2 , to the solution of the non-abelian theory [10] and also the obtention of correlation functions for the CGN model.

Recently, a functional integral approach has been developed to study gauge theories coupled to massless fermions in two dimensions. It has lead to the exact solution of the abelian model in a very simple way [11] and to new insights into the still unsolved non-abelian case [12, 13]. Basically, this approach parallels, in the path-integral framework, the operator fit of Lowenstein and Swieca [6] through the use of a chiral change in the fermionic variables.

It is the purpose of this work to extend this treatment to 2-dimensional self-interacting fermion models in order to study some of the unsolved questions mentioned before. In fact, one of the advantages of the approach is that, based on the introduction of auxiliary vector fields, it allows the establishment of a close parallel between purely fermionic models and gauge theories coupled to fermions.

Moreover, the auxiliary fields have the important rôle of displaying the internal group structure in a clear form. In particular, we shall be able to show the factorization of the $U(1)$ part of the CGN model at the level of Green functions, a point which was not completely clear using the hamiltonian formulation based on a plane wave type basis for the Bethe ansatz. It is also interesting to note that the auxiliary fields may allow one to make contact with a recent formulation of two-dimensional field theories developed by Marino, Schroer and Swieca [14] based on order-disorder variables within the path-integral framework.

The plan of the paper is as follows: In sect. 2 we define the models we shall study and show how the introduction of auxiliary vector fields allows for a natural extension of the techniques used in QED_2 and QCD_2 to the case of self-interacting fermion models.

Sect. 3 is devoted to the study of the Thirring model and shows how our approach leads to the spin $\frac{1}{2}$ Klaiber's solution [5] in a straightforward way, parallel to that leading to the solution of the Schwinger model in ref. [10].

For the chiral invariant Gross-Neveu model, discussed in sect. 4, we were not able to compute the correlation functions in a closed form – in fact we are stopped by the same problems preventing a complete solution in QCD_2 [12, 13]. However, we do gain new insight into some relevant features of the model. In particular, the property of factorization of the $U(1)$ part of the model can be clearly seen at the level of correlation functions, this being related to Witten's [16] prediction on dynamical

mass generation without spontaneous symmetry breaking (confirmed at the spectral level by Andrei and Lowenstein [9]).

In sect. 5 we give a brief summary of our results and conclusions. Finally, an appendix is added to sketch the computation of the jacobian arising from the chiral transformations of fermionic variables.

2. The auxiliary vector field

We shall consider two-dimensional fermion models with dynamics determined by the (euclidean) lagrangian density

$$\mathcal{L} = -i\bar{\psi}\not{\partial}\psi + \mathcal{L}_{\text{int}}(\bar{\psi}, \psi), \quad (2.1)$$

with

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \quad \gamma_5 = i\gamma_0\gamma_1, \quad \gamma_\mu\gamma_5 = i\epsilon_{\mu\nu}\gamma_\nu, \quad \epsilon_{01} = 1.$$

We shall study different interaction lagrangians. First, the simplest one corresponding to the Thirring model,

$$\mathcal{L}_{\text{int}} = -\frac{1}{2}g^2(\bar{\psi}\gamma_\mu\psi)^2, \quad (2.2)$$

invariant under the global abelian and abelian chiral groups:

$$\begin{aligned} \psi &\rightarrow e^{i\alpha}\psi, \\ \psi &\rightarrow e^{\gamma_5\beta}\psi, \end{aligned} \quad \alpha, \beta \in \mathbb{R}.$$

Next, we shall consider the case in which the fermions belong to the fundamental representation of $SU(N)$. Particularly, we are interested in the chiral invariant Gross-Neveu model,

$$\mathcal{L}_{\text{int}} = -\frac{g^2}{4N}[(\bar{\psi}_k\psi_k)^2 - (\bar{\psi}_k\gamma_5\psi_k)^2], \quad (2.3)$$

invariant under global $U(1) \times U(1)$ and also under global $SU(N)$ transformations

$$\psi \rightarrow \Omega\psi, \quad \Omega \in SU(N).$$

A Fierz type transformation can be used to write the interaction lagrangian (2.3) in terms of the $U(N)$ generators:

$$\mathcal{L}_{\text{int}} = -\frac{g^2}{2N}(\bar{\psi}\gamma_\mu\lambda^a\psi)^2, \quad a = 0, 1, \dots, N^2 - 1. \quad (2.4)$$

where

$$\begin{aligned}\lambda^0 &= \tfrac{1}{2}I, \\ \lambda^j &= t^j, \quad j = 1, 2, \dots, N^2 - 1,\end{aligned}\tag{2.5}$$

t_j being the $SU(N)$ generators normalized according to

$$\text{tr}^c(t^i t^j) = \tfrac{1}{2}\delta^{ij}.$$

It is interesting to note that when written in the form (2.4), the γ_5 matrices do not appear explicitly in the interaction. As we shall see, it is through the change of variables that we perform in the fermion degrees of freedom that they come into play, and the auxiliary vector fields to be introduced below are precisely at the heart of the process of keeping track of both γ_5 and $SU(N)$ symmetries properly.

It will become useful for our purposes to also study a simplified version of the interaction lagrangian (2.4), defined by restricting the t generators to those generating the maximal abelian subgroup (the torus) of $SU(N)$, which we shall call $SU(N)_D$. The interaction lagrangian is in this case given by

$$\mathcal{L}_{\text{int}} = -\frac{g^2}{8N}(\bar{\psi}\gamma_\mu\psi)^2 - \frac{g^2}{2N}(\bar{\psi}\gamma_\mu t^d\psi)^2\tag{2.6}$$

where t^d are the generators of $SU(N)_D$ ($d = 1, 2, \dots, N - 1$).

Notice that eq. (2.6) cannot be expressed, through Fierz transformations, in terms of γ_5 interactions as was the case for the complete model (2.4). However, as it happens for the QCD_2 case, we hope that some of the phenomena encountered in the $SU(N)_D$ model should extend to the complete $SU(N)$ case.

Let us now discuss the introduction of the auxiliary vector fields. In the path-integral formulation of the self-interacting fermion models, we simply make use of an identity which for single component fermions takes the form:

$$\exp\left\{+\tfrac{1}{2}g^2\int(\bar{\psi}\gamma_\mu\psi)^2d^2x\right\} = \int \mathcal{D}A_\mu \exp\left\{-\int\left[-g\bar{\psi}A_\mu\gamma^\mu\psi + \tfrac{1}{2}A_\mu A^\mu\right]d^2x\right\},\tag{2.7}$$

where A_μ is an abelian vector field. For the case of N -component fermions with interactions of the form (2.4) or (2.6) the natural fields to introduce are, apart for an

abelian vector field a non-abelian one, namely:

$$\exp\left\{\frac{g^2}{2N}\int(\bar{\psi}\gamma_\mu\lambda^a\psi)^2d^2x\right\}=\int\mathcal{D}A_\mu^0\mathcal{D}A_\mu\exp\left\{-\int\left[-\frac{g}{\sqrt{N}}\bar{\psi}\gamma_\mu\left(A_\mu\cdot\boldsymbol{t}+A_\mu^0\frac{I}{2}\right)\psi\right.\right. \\ \left.\left.+\frac{1}{2}A_\mu\cdot A^\mu+\frac{1}{2}A_\mu^0A^{0\mu}\right]d^2x\right\}, \quad (2.8)$$

where $A_\mu = A_\mu \cdot \boldsymbol{t}$ belongs to the Lie algebra of $SU(N)$ [or $SU(N)_D$ for the interaction lagrangian (2.6)].

At this point it is interesting to notice that keeping the CGN lagrangian in the form (2.3), the auxiliary fields that eliminate the quartic interaction have to be scalars, either for 1-color component fermions or N -component ones. As originally shown by Gross and Neveu [2] these fields are helpful to explore global aspects of the theory and to perform $1/N$ expansions: in this way relevant aspects of the theory like dynamical mass generation and asymptotic freedom were discovered. However, as we shall see below, some features where the symmetry plays an important rôle (for example the realization of the chiral symmetry [16]) can be more clearly discussed in a framework where lagrangians with one-component and N -component fermions, with interactions of the form (2.2) and (2.4) respectively are not treated on the same footing.

We now illustrate the details of our approach by considering the simplest case of the Thirring model. The generating functional for the Green functions is given by

$$Z[\bar{\theta}, \theta] = \mathcal{N} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left\{-\int\left[\bar{\psi}(-i\partial\psi) - \frac{1}{2}g^2(\bar{\psi}\gamma^\mu\psi)^2\right]d^2x\right\} \\ \times \exp\left\{\int[\bar{\psi}\theta + \bar{\theta}\psi]d^2x\right\}. \quad (2.9)$$

After the use of eq. (2.7) it becomes

$$Z[\bar{\theta}, \theta] = \mathcal{N} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}A_\mu \exp\left\{-\int\mathcal{L}'(A_\mu, \bar{\psi}, \psi)d^2x\right\} \exp\left\{\int[\bar{\theta}\psi + \bar{\psi}\theta]d^2x\right\}, \quad (2.10)$$

where

$$\mathcal{L}'(A_\mu, \bar{\psi}, \psi) = -\bar{\psi}(i\partial + gA)\psi + \frac{1}{2}A_\mu A^\mu. \quad (2.11)$$

We are then faced to an effective lagrangian with fermions coupled canonically to A_μ . This vector field carries mass dimensions and has no kinetic term. Moreover $\mathcal{L}'$ is

not gauge invariant and in fact A_μ contains two degrees of freedom. In spite of these differences, exactly as happens in the true gauge theory (the Schwinger model) A_μ can be decoupled from the fermions by making a change of variables [10] which takes the form

$$\begin{aligned}\psi(x) &= e^{i\eta(x) + \gamma_5 \phi(x)} \chi(x), \\ \bar{\psi}(x) &= \bar{\chi}(x) e^{-i\eta(x) + \gamma_5 \phi(x)};\end{aligned}\quad (2.12)$$

$$A_\mu(x) = -\frac{1}{g}(\epsilon_{\mu\nu} \partial_\nu \phi(x) - \partial_\mu \eta(x)). \quad (2.13)$$

Here the transformations (2.12) factorize the U(1) part of the fermion fields and at the same time identifies it with A_μ through the change (2.13). Hence, the new variables are now the fermions χ and $\bar{\chi}$ and the scalars ϕ and η . In terms of these new variables $\mathcal{L}'$ is written as

$$\mathcal{L}'(\bar{\chi}, \chi, \phi, \eta) = \left\{ -\bar{\chi} i \not{\partial} \chi + \frac{1}{2g^2} \partial_\mu \phi \partial^\mu \phi + \frac{1}{2g^2} \partial_\mu \eta \partial^\mu \eta + \frac{1}{g} \epsilon_{\mu\nu} \partial_\mu \phi \partial_\nu \eta \right\}, \quad (2.14)$$

where χ and $\bar{\chi}$ are completely decoupled from the rest.

As has been shown by Fujikawa [17], the transformation (2.12) is not free of cost due to the non-invariance of the fermionic measure under a chiral change of variables. The non-triviality of the resulting jacobian is directly related to the Atiyah-Singer index theorem and it is in this context that topology plays a rôle in our formulation of purely fermionic models (see refs. [12, 17]). The contribution of this jacobian adds to (2.14) a term which depends on ϕ giving an effective lagrangian which will be discussed in the next sections.

Concerning the vector field, transformation (2.13) is easily seen to give the following jacobian:

$$\mathcal{D}A_0 \mathcal{D}A_1 = \det \left[-\frac{1}{g^2} (\partial_0^2 + \partial_1^2) \right] \mathcal{D}\phi \mathcal{D}\eta. \quad (2.15)$$

One can recognize the Faddeev-Popov determinant for the Lorentz gauge and this is not a coincidence since in fact (2.13) expresses the fact that the vector field has been decomposed into longitudinal and transverse parts.

We then see that the problem of solving the Thirring model has been related to one closely connected with the Schwinger model. As we shall see, a similar relation can be established between QCD₂ with massless fermions and the chiral invariant Gross-Neveu model. One faces in this last case the problem of handling a non-abelian generalization of transformations (2.12), (2.13) and hence the difficulties which are

characteristic of QCD₂ and have prevented, up to now, a closed solution for non-abelian gauge theories with massless fermions in two dimensions.

3. Correlation functions for the massless Thirring model

In order to compute Green functions for the Thirring model we start from the generating functional $Z[\bar{\theta}, \theta]$ in the form (2.10) and make the change of variables (2.12)–(2.13).

$$\begin{aligned} Z[\bar{\theta}, \theta] = & \mathcal{N} \int \mathcal{D}\bar{\chi} \mathcal{D}\chi \mathcal{D}\phi \mathcal{D}\eta J_F J_A \exp \left\{ - \int [-\bar{\chi} i \not{\partial} \chi] d^2x \right\} \\ & \times \exp \left\{ \frac{1}{2g^2} \int [(\partial_\mu \phi)^2 + (\partial_\mu \eta)^2] d^2x \right\} \exp \left\{ \int [\bar{\theta} e^{\gamma_5 \phi + i\eta} \chi + \bar{\chi} e^{\gamma_5 \phi - i\eta} \theta] d^2x \right\} \end{aligned} \quad (3.1)$$

where J_F and J_A are the jacobians of the fermion transformation (2.12) and the vector transformation (2.13) respectively, $\mathcal{N}$ being a normalization factor.

The factor J_A is given by eq. (2.15) and, being constant, it can be absorbed in $\mathcal{N}$. Concerning J_F , the computation of jacobians arising from chiral changes in the fermionic variables has been performed in detail in refs. [11–13]. We then only sketch the principal steps in its derivation in the appendix. From eq. (A.32) we have:

$$J_F = \exp \left\{ \frac{1}{2\pi} \int d^2x [\phi \partial_\mu \partial^\mu \phi + \epsilon_{\mu\nu} \phi \partial_\mu \partial_\nu \eta] \right\}. \quad (3.2)$$

Hence, $Z[\bar{\theta}, \theta]$ is finally written as

$$\begin{aligned} Z[\bar{\theta}, \theta] = & \mathcal{N} \int \mathcal{D}\bar{\chi} \mathcal{D}\chi \mathcal{D}\phi \mathcal{D}\eta \exp \left\{ - \int \mathcal{L}_{\text{eff}} d^2x \right\} \\ & \times \exp \left\{ \int [\bar{\theta} e^{\gamma_5 \phi + i\eta} \chi + \bar{\chi} e^{\gamma_5 \phi - i\eta} \theta] d^2x \right\}, \end{aligned} \quad (3.3)$$

with

$$\mathcal{L}_{\text{eff}} = -\bar{\chi} i \not{\partial} \chi + \frac{1}{2g^2} \left(1 + \frac{g^2}{\pi} \right) (\partial_\mu \phi)^2 + \frac{1}{2g^2} (\partial_\mu \eta)^2. \quad (3.4)$$

We have neglected in (3.4) two types of contributions to the effective lagrangian:

- (i) Those coming from terms of the form $\epsilon_{\mu\nu} \partial_\mu \partial_\nu \phi$ or $\epsilon_{\mu\nu} \partial_\mu \partial_\nu \eta$.
- (ii) Those originated by the integration by parts which leads to (3.4). Namely, we have used repeatedly

$$\int \partial_\mu (\phi \partial_\mu \phi) d^2x = 0. \quad (3.5)$$

Under these restrictions we get a very simple effective lagrangian which corresponds to two free scalar fields and, as we shall see below, the analysis of fermion Green functions will be straightforward.

The price we have to pay to achieve such simplicity is to exclude from the domain of path integration some configurations carrying non-zero winding number which, *a priori*, should be important at the level of Green functions. Our treatment is then *exact only in the zero winding number sector*. In support of our approximation we can mention that, in refs. [11,12] the same procedure did not spoil the physical information* in the sense that all Green functions and composite quantum matrix elements relevant for the computation of any structure function can be derived in complete agreement with the results obtained using bosonization techniques [6]. In what follows, we show that the same is true for the Thirring model; namely, we are able to reproduce the *exact* Klaiber solution for spin $\frac{1}{2}$. Notice that the intrinsic mechanism that originated the mass term in the Schwinger model generates the shift in the constant in front of $(\partial_\mu \phi)^2$ with respect to that in front of $(\partial_\mu \eta)^2$; as we shall see, this change is very important at the level of Green functions.

It is now straightforward to compute any correlation function. For example the two-point function is given by:

$$\begin{aligned} \langle \psi(x) \bar{\psi}(0) \rangle &= \frac{1}{\mathcal{N}} \left. \frac{\delta^2 Z}{\delta \bar{\theta}(x) \delta \theta(0)} \right|_{\theta=0=\bar{\theta}} \\ &= \langle e^{i\eta(x) + \gamma_5 \phi(x)} \chi(x) \bar{\chi}(0) e^{-i\eta(0) + \gamma_5 \phi(0)} \rangle \\ &= \langle \chi(x) \bar{\chi}(0) \rangle_0 \langle e^{i[\eta(x) - \eta(0)] + \gamma_5 [\phi(x) - \phi(0)]} \rangle, \end{aligned} \quad (3.6)$$

where $\langle \chi(x) \bar{\chi}(0) \rangle_0$ is the free fermion Green function

$$\langle \chi(x) \bar{\chi}(0) \rangle_0 = \text{const} \cdot \frac{\gamma_\mu x^\mu}{x^2}. \quad (3.7)$$

In order to calculate the second factor in (3.6) we use the expression for the massless scalar Green function:

$$D(x) = -\frac{1}{2\pi} \log \mu |x| \quad (3.8)$$

satisfying $\square^2 D(x) = -\delta^{(2)}(x)$.

*In fact, for the Schwinger model, it has been shown in refs. [23, 26] under a careful treatment that the kind of contribution we are neglecting in the effective lagrangian (3.4) (which comes from the non-zero winding sector) is exactly cancelled by the one arising from zero eigenmodes in the computation of Green functions.

Then, calling

$$\alpha^2 = \frac{1}{g^2}, \quad \beta^2 = \left(\frac{1}{g^2} + \frac{1}{\pi} \right), \quad (3.9)$$

and ignoring all terms involving contraction of fields at the same point (see ref. [18]) we finally have

$$\langle \psi(x) \bar{\psi}(0) \rangle = \exp \left\{ -\frac{1}{2\pi} \left(\frac{1}{\alpha^2} - \frac{1}{\beta^2} \right) \log \mu |x| \right\} \langle \chi(x) \bar{\chi}(0) \rangle_0. \quad (3.10)$$

The result above can be shown to be the spin $\frac{1}{2}$ Klaiber solution up to a Klein transformation and a redefinition of the coupling constant. One should easily extend this result to general n -point correlation functions.

It is worthwhile to notice that it is the contribution of J_F that prevents (3.10) from being just the free fermion Green function, since it is due to its presence that $\alpha^2 \neq \beta^2$. As stressed by Fujikawa [17] the nontriviality of J_F is related to the Atiyah-Singer index theorem and it is in this sense that topology plays an important rôle in the study of self-interacting fermion models. Hidden in the usual treatments of purely fermionic models, it becomes apparent in our approach and, on this basis, some comments are in order concerning some not completely solved problems in the Bethe ansatz approach to the Thirring model (or the Luttinger model [19]). In particular, there is the question of why a Bethe ansatz wave function fails to give [19,20] the correct correlation functions, whereas the operator construction or bosonization techniques lead to the correct answer. In our path-integral approach, the adequate basis to expand the fermion variables [17,21] taking account of topological effects is not the plane-wave basis but the one defined by the eigenvalue equation

$$\not{D}\varphi_n = \lambda_n \varphi_n, \quad (3.11)$$

where $\not{D}$ is the covariant derivative defined in terms of A_μ . The chiral asymmetry appears in this basis in the well-defined zero-eigenvalue sector but it is transferred to the infinite frequencies sector for plane waves. This is the reason why a naive computation in the plane-wave basis gives a trivial and wrong result for the jacobian. The Bethe ansatz is precisely based in the plane-wave basis and hence, unless the above problems are properly treated, it cannot give the right answer in the cases where the chiral asymmetry plays an important rôle*.

In the solution of Luttinger model given by Mattis and Lieb [22], this problem seems to be overcome by filling the negative energy sea of spinless fermions up to

*The authors of ref. [9] overcome this problem by adding to the original lagrangian a vector current term which is used at the end of the calculations to subtract the Thirring-like excitations.

infinite frequency and the manifestation of chiral asymmetry appears in the form of non-commutativity of certain bosonic operators. Since the Bethe ansatz method has been so fruitful in solving many problems [8, 9], we think this is an important aspect to be investigated in detail.

4. The chiral invariant Gross-Neveu model

We turn now to the case of the chiral invariant Gross-Neveu model written in the form (2.4), namely

$$\mathcal{L}_{\text{int}} = -\frac{g^2}{N} (\bar{\psi} \gamma_\mu \lambda^a \psi)^2. \quad (4.1)$$

The introduction of auxiliary vector yields using identity (2.8) allows us to make a parallel between the present model and QCD_2 . In order to explore this parallel let us review the situation in QCD_2 , a theory where there have been some attempts to compute fermionic Green functions using functional methods [12, 13, 23]. In ref. [12] a transformation decoupling fermions from gluons was introduced and the corresponding jacobian was computed. However, it was not possible to write the jacobian as a local function of the gluon fields, neither was clear if a gauge condition was implicit in the change of variables. More recently, Roskies [13] was able to answer positively to this last question but the problem of expressing the gauge condition in a differential form remains unsolved [although much progress has been made by Roskies for the $\text{SU}(2)$ case]. Then, the computation of the Faddeev-Popov determinant cannot be performed and there is no proof that a complete intrinsic Higgs mechanism [24] takes place in QCD_2 as happens in the Schwinger model and the $\text{SU}(N)_D$ theory [3, 10, 12].

Let us now make more precise the statement above that most of the difficulties involved in solving the self-interacting fermion models with $\mathcal{L}_{\text{int}}$ given by eq. (4.1) are similar to those one encounters in QCD_2 with massless fermions.

We start from the generating functional for the CGN model with the interaction lagrangian written in the form (4.1). Through the use of eq. (2.8) we can express it in terms of auxiliary fields A_μ^a ($a = 0, 1, \dots, N^2 - 1$) as follows:

$$Z[\bar{\theta}, \theta] = \mathcal{N} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}A_\mu^a \exp \left\{ - \int [\mathcal{L}'(A_\mu^a, \bar{\psi}, \psi)] d^2x \right\} \exp \left\{ \int [\bar{\theta}\psi + \bar{\psi}\theta] d^2x \right\}, \quad (4.2)$$

where

$$\mathcal{L}'(A_\mu^a, \bar{\psi}, \psi) = -\bar{\psi} \not{D} \psi + \frac{1}{2} (A_\mu \cdot A_\mu + A_\mu^0 A_\mu^0). \quad (4.3)$$

Here A_μ^0 is an abelian vector field whereas A_μ is a $2(N^2 - 1)$ -component vector. The covariant derivative is given by

$$D_\mu = i\partial_\mu + \frac{g}{\sqrt{N}} \left(A_\mu^0 \frac{I}{2} + A_\mu \cdot \mathbf{t} \right). \quad (4.4)$$

We can extract the U(1) part of the fermionic variables by considering an abelian chiral transformation, already familiar from sect. 2; the abelian vector field will be then decoupled from the fermions. Namely, we perform the following change of variables:

$$\psi(x) = e^{i\eta_0(x) + \gamma_5 \phi_0(x)} \chi(x) \equiv U_0 \chi(x), \quad (4.5)$$

$$\bar{\psi}(x) = \bar{\chi}(x) e^{-i\eta_0(x) + \gamma_5 \phi_0(x)} \equiv \bar{\chi}(x) U_0^* ;$$

$$A_\mu^0 = -\frac{\sqrt{N}}{g} (\epsilon_{\mu\nu} \partial_\nu \phi_0 - \partial_\mu \eta). \quad (4.6)$$

In terms of the new variables $\bar{\chi}, \chi, \phi^0, \eta^0$, eq. (4.2) becomes

$$\begin{aligned} Z[\bar{\theta}, \theta] = & \mathcal{N} \int \mathcal{D}\bar{\chi} \mathcal{D}\chi \mathcal{D}A_\mu \mathcal{D}\phi^0 \mathcal{D}\eta^0 J_{A^0} J_{U(1)}^F \exp \left\{ - \int \left[-\bar{\chi} \not{D}' \chi + \frac{1}{2} A_\mu \cdot A_\mu \right] d^2x \right\} \\ & \times \exp \left\{ - \frac{N}{2g^2} \int d^2x \left[(\partial_\mu \phi_0)^2 + (\partial_\mu \eta_0)^2 \right] \right\} \\ & \times \exp \left\{ \int \left[\bar{\theta} e^{\gamma_5 \phi^0 + i\eta^0} \chi + \bar{\chi} e^{\gamma_5 \phi^0 - i\eta^0} \theta \right] d^2x \right\}, \end{aligned} \quad (4.7)$$

where

$$D'_\mu = i\partial_\mu + \frac{g}{\sqrt{N}} (A_\mu \cdot \mathbf{t}). \quad (4.8)$$

The jacobian J_{A^0} is the trivial one given by eq. (3.6) while the fermionic one, $J_{U(1)}^F$, is

$$J_{U(1)}^F = \exp \left\{ - \frac{N}{2\pi} \int d^2x (\partial_\mu \phi^0)^2 \right\}. \quad (4.9)$$

We have again neglected surface terms and hence our treatment will be exact only in the zero winding-number sector (see the discussion in sect. 3).

The generating functional can then be written as

$$\begin{aligned}
 Z[\bar{\theta}, \theta] = & \mathcal{N} \int \mathcal{D}\bar{\chi} \mathcal{D}\chi \mathcal{D}\mathbf{A}_\mu \mathcal{D}\phi^0 \mathcal{D}\eta^0 \\
 & \times \exp \left\{ - \int \left[-\bar{\chi} \not{D}' \chi + \frac{1}{2} \mathbf{A}_\mu \cdot \mathbf{A}_\mu + \frac{N}{2g^2} \left(1 + \frac{g^2}{\pi} \right) (\partial_\mu \phi_0)^2 + \frac{N}{2g^2} (\partial_\mu \eta^0)^2 \right] d^2x \right\} \\
 & \times \exp \left\{ \int \left[\bar{\theta} e^{\gamma_5 \phi^0 + i\eta^0} \chi + \bar{\chi} e^{\gamma_5 \phi^0 - i\eta^0} \theta \right] d^2x \right\}. \quad (4.10)
 \end{aligned}$$

Next, we shall consider a non-abelian chiral transformation involving the generators of the $SU(N)$ group:

$$\begin{aligned}
 \chi(x) &= U^{-1}(x) U_5(x) \xi(x), \\
 \bar{\chi}(x) &= \bar{\xi}(x) U_5(x) U(x), \quad (4.11)
 \end{aligned}$$

and

$$\gamma_\mu \mathbf{A}_\mu \cdot \mathbf{t} = U^{-1} \left(\frac{i\sqrt{N}}{g} \not{\partial} U_5 \right) U_5^{-1} U - U^{-1} \left(\frac{i\sqrt{N}}{g} \not{\partial} U \right), \quad (4.12)$$

where

$$U(x) = e^{it \cdot \eta(x)}, \quad (4.13)$$

$$U_5(x) = e^{\gamma_5 t \cdot \Phi(x)}. \quad (4.14)$$

We are then introducing as new variables the fermionic fields $\xi, \bar{\xi}$ and $2(N^2 - 1)$ fields ϕ^a, η^a ($a = 1, 2, \dots, N^2 - 1$).

It is important to stress at this point that any vector field $\mathbf{A}_\mu$ can be written in the form (4.12). To see that this is true, we first note that eq. (4.12) expresses $\mathbf{A}_\mu$ as the gauge-transformed of a field $\mathbf{A}_\mu^g$ defined as

$$\gamma^\mu \mathbf{A}_\mu^g \cdot \mathbf{t} = \frac{i\sqrt{N}}{g} (\not{\partial} U_5) U_5^{-1}. \quad (4.15)$$

Now, Roskies has shown [13] by passing to a $j = i\gamma_5$ complexification ($j^2 = -1$) of the gauge group that one can always choose a gauge such that any two-dimensional gauge field can be written in the form (4.15). It is precisely in this gauge (the decoupling gauge) that QCD₂ fermions can be decoupled from the gluons exactly as happens in the abelian theory (in the Lorentz gauge).

It is then clear that any vector field A_μ can be expressed as the gauge transform of (4.15) and this is precisely the meaning of (4.12).

Under the change of variables defined by eqs. (4.11), (4.12) the lagrangian becomes

$$\begin{aligned} \mathcal{L}''(\bar{\xi}, \xi, \phi^a, \eta^a, \phi^0, \eta^0) = & \left\{ \bar{\xi}(-i\partial\xi) + \frac{N}{2g^2} \left(1 + \frac{g^2}{\pi}\right) (\partial_\mu \phi_0)^2 + \frac{N}{2g^2} (\partial_\mu \eta_0)^2 \right. \\ & \left. + \text{Tr} \left[U^{-1} \left(\frac{i\sqrt{N}}{g} \partial U_5 \right) U_5^{-1} U - \frac{i\sqrt{N}}{g} U^{-1} \partial U \right]^2 \right\}, \quad (4.16) \end{aligned}$$

or

$$\mathcal{L}'' = \bar{\xi}(-i\partial\xi) + \frac{N}{2g^2} \left(1 + \frac{g^2}{\pi}\right) (\partial_\mu \phi_0)^2 + \frac{N}{2g^2} (\partial_\mu \eta_0)^2 + \frac{1}{2} \text{Tr}(\gamma_\mu A_\mu \cdot t)^2. \quad (4.17)$$

Hence, the generating functional in terms of the new variables reads:

$$\begin{aligned} Z[\bar{\theta}, \theta] = & \mathcal{N} \int \mathcal{D}\xi \mathcal{D}\bar{\xi} \exp \left\{ - \int \bar{\xi}(-i\partial\xi) d^2x \right\} \mathcal{D}\eta^0 \mathcal{D}\phi^0 \mathcal{D}\eta^a \mathcal{D}\phi^a \\ & \times J_F^{\text{SU}(N)} J_A \exp \left\{ - \int \left[\frac{N}{2g^2} \left(1 + \frac{g^2}{\pi}\right) (\partial_\mu \phi_0)^2 + \frac{N}{2g^2} (\partial_\mu \eta_0)^2 \right] d^2x \right\} \\ & \times \exp \left\{ - \int \frac{1}{2} A_\mu(\phi_a, \eta_a) \cdot A_\mu(\phi^a, \eta^a) d^2x \right\} \\ & \times \exp \left\{ \int [\bar{\theta} U_0 U^{-1} U_5 \xi + \bar{\xi} U_5 U U_0^* \theta] d^2x \right\} \quad (4.18) \end{aligned}$$

Here $J_F^{\text{SU}(N)}$ represents the jacobian of the U_5 change in the fermionic variables. Note that the U -part of the transformation does not affect the fermionic integration measure (which is defined in a gauge-invariant way). The computation of $J_F^{\text{SU}(N)}$ can be performed following the steps of the QCD₂ case (see the appendix). Concerning the change in the vector fields

$$\mathcal{D}A_0^a \mathcal{D}A_1^a = J_A \mathcal{D}\phi^a \mathcal{D}\eta^a, \quad a = 1, 2, \dots, N^2 - 1, \quad (4.19)$$

which contributed with a constant jacobian in the abelian case, it produces now a very complicated jacobian (which coincides with the Faddeev-Popov determinant in the decoupling gauge).

However, even without computing $J_F^{\text{SU}(N)}$ and J_A one can already see a remarkable property of the model: first, from the generating functional (4.18) one can see that

the massless excitations coming from the U(1) charge decouple from the rest, as predicted by Witten [16] and confirmed at the spectral level by Andrei and Lowenstein [9]. Moreover, in agreement with the work of Rothe and Swieca [25] we can show that this part of the Green functions, that indeed factorizes, coincides with the Thirring model U(1) part computed in the previous section. To see all these properties let us compute the two-point correlation function from the generating functional (4.18):

$$\begin{aligned}
 \langle \psi_l(x) \bar{\psi}_{l'}(0) \rangle &= \mathcal{N}^{-1} \left. \frac{\delta^2 Z[\bar{\theta}, \theta]}{\delta \bar{\theta}_l(x) \delta \theta_{l'}(0)} \right|_{\bar{\theta}=0=\theta} \quad (l, l' = 1, 2, \dots, N) \\
 &= \int \mathcal{D}\phi^0 \mathcal{D}\eta^0 U_0(\phi^0, \eta^0) U_0^*(\phi^0, \eta^0) \\
 &\quad \times \exp \left\{ - \int \left[\frac{N}{2\beta^2} (\partial_\mu \phi^0)^2 + \frac{N}{2\alpha^2} (\partial_\mu \eta^0)^2 \right] d^2x \right\} \\
 &\quad \times \int \mathcal{D}\bar{\xi} \mathcal{D}\xi \xi_k(x) \xi_{k'}(0) \exp \left\{ - \int d^2x \bar{\xi} (-i \not{\partial} \xi) \right\} \\
 &\quad \times \int \mathcal{D}\phi^a \mathcal{D}\eta^a J_F^{\text{SU}(N)} J_A (U^{-1}(x) U_5^{-1}(x))_{kl} (U(0) U_5(0))_{k'l'} \\
 &\quad \times \exp \left\{ - \int \frac{1}{2} A_\mu(\phi^a, \eta^a) \cdot A_\mu(\phi^a, \eta^a) d^2x \right\}, \quad (4.20)
 \end{aligned}$$

or, shortly,

$$\begin{aligned}
 \langle \psi_l(x) \bar{\psi}_{l'}(0) \rangle &= \langle U_0(x) U_0^*(0) \rangle_{\text{U}(1)} \langle \xi_k(x) \bar{\xi}_k(0) \rangle_0 \\
 &\quad \times \langle (U^{-1}(x) U_5^{-1}(x))_{kl} (U(0) U_5(0))_{k'l'} \rangle, \\
 &\quad l, l' = 1, 2, \dots, N. \quad (4.21)
 \end{aligned}$$

The first factor corresponds to the U(1) part of the Green function and it is similar to the one appearing in the Thirring model:

$$\langle U_0(x) U_0^*(0) \rangle_{\text{U}(1)} = (\mu |x|)^{-\kappa(g)/N}, \quad (4.22)$$

where

$$\kappa(g) = \frac{g^2}{2\pi} \left(1 - \frac{1}{1 + g^2/\pi} \right) \geq 0. \quad (4.23)$$

We can then infer that the “almost long-range order” of the Kosterlitz-Thouless type occurs [16] in the infrared region (dominated by the massless particles). On the other hand, for short distances, the coupling constant goes to zero (due to asymptotic freedom) the dynamically generated mass remains and the chirality carrying term (4.22) goes to 1 ensuring that in the asymptotically free region the massless part is not important.

It is the third factor in eq. (4.21) which is the one presenting difficulties. Let us consider for simplicity the SU(2) case and write explicitly the different quantities appearing in it. The transformations U and U_5 can be written for SU(2) as

$$\begin{aligned} U(x) &= \cos \eta(x) + i\sigma \cdot \hat{\eta} \sin \eta(x), \quad \hat{\eta} = \eta/|\eta|, \quad \eta = \frac{1}{2}|\eta|, \\ U_5(x) &= \cosh \phi(x) + \gamma_5 \sigma \cdot \hat{\phi} \sinh \phi(x), \end{aligned} \quad (4.24)$$

and the expression for A_μ in terms of ϕ and η is given by

$$A_\mu = B_\mu + C_\mu + \frac{1}{2} \left[(\hat{\eta} \wedge B_\mu) \sin \eta + 2((\hat{\eta} \cdot B_\mu) \hat{\eta} - B_\mu) \sin^2 \eta \right], \quad (4.25)$$

with B_μ and C_μ given by

$$B_\mu = -\frac{\sqrt{N}}{g} \left\{ \varepsilon_{\mu\nu} [\hat{\phi} \partial_\nu \phi + \frac{1}{2} (\partial_\nu \hat{\phi}) \sinh \phi] - (\hat{\phi} \wedge \partial_\mu \hat{\phi}) \sinh^2 \phi \right\}, \quad (4.26)$$

$$C_\mu = -\frac{\sqrt{N}}{g} \left\{ \hat{\eta} \partial_\mu \eta + (\partial_\mu \hat{\eta}) \sin \eta + (\hat{\eta} \wedge \partial_\mu \hat{\eta}) 2 \sin^2 \eta \right\}. \quad (4.27)$$

In order to compute the jacobian $J_F^{\text{SU}(N)}$ one proceeds as in the appendix getting

$$J_F^{\text{SU}(N)} = \exp \left\{ -\frac{\varepsilon_{\mu\nu}}{\pi} \int d^2x \int_0^1 d\alpha \phi \cdot F_{\mu\nu}(\alpha) \right\}, \quad (4.28)$$

where

$$F_{\mu\nu}(\alpha) = \partial_\mu A_\nu(\alpha) - \partial_\nu A_\mu(\alpha) + 2A_\mu(\alpha) \wedge A_\nu(\alpha), \quad (4.29)$$

$$A(\alpha) = A + \frac{\sqrt{N}}{g} U_5(\alpha) \not{D} U_5(\alpha). \quad (4.30)$$

For the QCD₂ theory, working in the decoupling gauge, $C_\mu = 0$ and it is possible to perform the α -integration [13]. This leads to a non-linear local interaction in terms of ϕ . In the present case there is the additional complication of the η degrees of freedom which, *a priori*, play a rôle as important as that of ϕ ; we have not been able to perform the integration and hence the analysis of the SU(N) part of the correlation functions becomes very complicated. For example, the quadratic term

$A_\mu \cdot A_\mu$ takes the form:

$$\frac{1}{2} A_\mu \cdot A_\mu = \frac{1}{8g^2} \text{Tr} \{ \mathcal{B} \mathcal{B} + [\mathcal{B} U, \mathcal{C} U] + 2\mathcal{C} U^{-1} \mathcal{B} U^{-1} + \mathcal{C} \mathcal{C} \}. \quad (4.31)$$

Then, up to this point we have learned that the fermions can be decoupled from the vector fields as in the QCD₂ case, the massless excitations decouple from the rest but the precise behavior of the SU(N) factor has not been determined due to difficulties peculiar to the non-abelian character of the model. We think however that some features of the color part can be inferred from a simplified theory. In fact, guided from the experience in QCD₂ [10, 12] where some of the features of the exact solution of the SU(N)_D theory seem to be present in the complete SU(N) case, we devote the last part of this section to the discussion of a model with interaction lagrangian (2.6):

$$\mathcal{L}_{\text{int}} = -\frac{g^2}{8N} (\bar{\psi} \gamma_\mu \psi)^2 - \frac{g^2}{2N} (\bar{\psi} t^d \gamma_\mu \psi)^2, \quad d = 1, 2, \dots, N-1, \quad (4.32)$$

t^d being the $N-1$ generators of SU(N)_D. The functional approach to the model proceeds exactly as for the SU(N) case except that now the change of variables (4.12) for the vector field takes the very simple form

$$A_\mu = -t^d \frac{\sqrt{N}}{g} [\epsilon_{\mu\nu} \partial_\nu \phi^d - \partial_\mu \eta^d], \quad d = 1, 2, \dots, N-1. \quad (4.33)$$

This, together with the change in the fermionic variables:

$$\begin{aligned} \chi_d(x) &= e^{i\eta^d(x) + \gamma_5 \phi^d(x)} \xi_d(x), \\ \bar{\chi}_d(x) &= \bar{\xi}_d(x) e^{-i\eta^d(x) + \gamma_5 \phi^d(x)}, \end{aligned} \quad (4.34)$$

gives a very simple effective lagrangian since the jacobian J_A for the vector field transformation is trivial while that arising from the fermionic one is simply given by (see the appendix):

$$J_F^{\text{SU}(N)_D} = \exp \left\{ -\frac{1}{4\pi} \int d^2x [\phi^d \partial_\mu \partial^\mu \phi^d + \epsilon_{\mu\nu} \phi^d \partial_\mu \partial_\nu \eta^d] \right\}. \quad (4.35)$$

Hence, the generating functional takes the form:

$$\begin{aligned} Z[\bar{\theta}, \theta] &= \mathcal{N} \int \mathcal{D}\bar{\xi} \mathcal{D}\xi \mathcal{D}\phi^0 \mathcal{D}\eta^0 \mathcal{D}\phi^d \mathcal{D}\eta^d \exp \left\{ -\int \mathcal{L}_{\text{eff}} d^2x \right\} \\ &\times \exp \left\{ \int \left[\bar{\theta} e^{i(\eta^0 + \eta^d t_d) + \gamma_5(\phi^0 + \phi^d t_d)} \xi + \bar{\xi} e^{-i(\eta^0 + \eta^d t_d) + \gamma_5(\phi^0 + \phi^d t_d)} \theta \right] d^2x \right\}, \end{aligned} \quad (4.36)$$

where $\mathcal{L}_{\text{eff}}$ is given by

$$\begin{aligned} \mathcal{L}_{\text{eff}} = & \bar{\xi}(-i\not{\partial}\xi) + \frac{N}{2g^2} \left(1 + \frac{g^2}{\pi}\right) (\partial_\mu \phi^0)^2 + \frac{N}{2g^2} (\partial_\mu \eta^0)^2 \\ & + \frac{N}{2g^2} \left[1 + \frac{g^2}{2N\pi}\right] (\partial_\mu \phi^d)^2 + \frac{N}{2g^2} (\partial_\mu \eta^d)^2. \end{aligned} \quad (4.37)$$

Here, we have again neglected surface terms.

One can easily compute any correlation function from eqs. (4.36), (4.37). For example, the two-point correlation function takes the form

$$\begin{aligned} \langle \psi(x) \bar{\psi}(0) \rangle = & \langle \xi(x) \bar{\xi}(0) \rangle_0 \exp \left\{ -\frac{g^2}{2\pi N} \left(1 - \frac{1}{1 + g^2/\pi}\right) \log \mu|x| \right\} \\ & \times \exp \left\{ -\frac{g^2(N-1)}{2\pi N} \left(1 - \frac{1}{1 + g^2/2N\pi}\right) \log \mu|x| \right\}, \end{aligned} \quad (4.38)$$

where the second factor corresponds to the $U(1)$ part and the third one to the $SU(N)_D$ one. It is clear that in this particular model there is no mass generation. There are N pairs of massless scalars $(\phi_0, \eta_0, \phi^d, \eta^d)$ determining the infrared behavior of the $U(1) \times SU(N)_D$ part, which may remain in the $U(1) \times SU(N)$ case. In this last theory it is known from the $1/N$ expansion that the fermions acquire a mass [2, 16].

5. Conclusions

We have analysed self-interacting fermion models in $d = 2$ dimensions using the functional integral method. The starting point of our treatment is the introduction of an auxiliary vector field, allowing us to establish a close connection between purely fermionic models with quartic interactions and 2-dimensional gauge theories with massless fermions.

Exploiting this relation, we applied the methods developed in QED_2 and QCD_2 theories [11, 12] (based on a change of variables which decouples the fermions from the gauge fields) to study the Thirring and the chiral invariant Gross-Neveu models.

For the case involving the $U(1)$ group, the Thirring model, we obtained the exact (spin $\frac{1}{2}$) solution. The auxiliary vector field is converted into two massless scalars whose propagators appear exponentiated in the Green functions, in agreement with the well-known Klaiber's solution. An important point which becomes apparent in our treatment is the rôle of topological properties in the behavior of the Green functions: It is clear from eq. (3.10) that the only thing responsible for the deviation

of the two-point correlation function behavior from a free fermion one is the non-triviality of the chiral transformation jacobian.

We then extended our approach to non-abelian theories and analysed the chiral invariant Gross-Neveu model by decoupling fermions from vector fields [now in the algebra of $U(N)$] performing a change of variables also useful for treating QCD_2 . In this way we were able to treat the chiral $U(1)$ charged massless part of the Green functions exactly, showing its factorization from the color part. Again the jacobian contribution is crucial and assures the almost long-range order predicted by Witten [16] in the infrared region. Our results also show the decoupling of color degrees of freedom from a free fermionic part in the two-point correlation function but we do not have an exact result for the $SU(N)$ factor – in fact we faced the same difficulties preventing a closed solution of QCD_2 . It is important to notice that in the short distance region, because of asymptotic freedom, the $U(1)$ factor which, as we showed has a power behavior depending on g (cf. with the g -independent $1/N$ result [16]) goes to 1.

We would like to note that our results – the solution of the Thirring model and the relevant features of the CGN one – were obtained by carefully treating the chiral aspects of the theory, in particular in the computation of the fermionic jacobian. It is then clear that, when using methods like the Bethe ansatz (which starts from a finite volume and is based in a plane-wave analysis) special care must be paid in the treatment of boundary conditions in order to correctly describe the chiral asymmetry of the theory.

As it has been stressed in sect. 3, our results were obtained neglecting surface terms which are non-zero if certain field configurations – carrying non-zero winding number – are to be considered. To take such configurations into account, the present method has to be extended possibly using techniques like the one developed in ref. [14], but this remains to be investigated further.

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Appendix

THE FERMION JACOBIAN

We shall compute the jacobian of the chiral transformation passing from the old fermion fields $\psi, \bar{\psi}$ to the new ones $\chi, \bar{\chi}$:

$$\begin{aligned}\psi(x) &= U_5(x)\chi(x), \\ \bar{\psi}(x) &= \bar{\chi}(x)U_5(x),\end{aligned}\tag{A.1}$$

$$U_5(x) = e^{\gamma_5\phi(x)},\tag{A.2}$$

where $\phi(x)$ is an element of the algebra of $SU(N)$. In terms of the new fermion fields the covariant derivative term appearing in the lagrangian (4.3) [or (2.11) for the abelian case]

$$\mathcal{L}_1(\psi, \bar{\psi}, A_\mu) = -\bar{\psi} \not{D} \psi = -\bar{\psi} \left(i \not{\partial} + \frac{g}{\sqrt{N}} \not{A} \right) \psi, \quad (\text{A.3})$$

is replaced by

$$\mathcal{L}_1(\chi, \bar{\chi}, A_\mu) = -\bar{\chi} \not{D}' \chi, \quad (\text{A.4})$$

where

$$D'_\mu = i \partial_\mu + \frac{g}{\sqrt{N}} A'_\mu, \quad (\text{A.5})$$

A'_μ being a shorthand notation for

$$A'_\mu = A_\mu + \frac{\sqrt{N}}{2g} \text{tr}(\gamma_\mu U_5 \not{D} U_5), \quad (\text{A.6})$$

where tr stands for the trace of γ -matrices products, tr^c will be used for the $SU(N)$ matrices and $\text{Tr} = \text{tr} \otimes \text{tr}^c$.

As usual, the functional integral fermionic measure is defined as [21]

$$\mathcal{D}\bar{\psi} \mathcal{D}\psi = \prod_{n,m} d\bar{b}_n da_m, \quad (\text{A.7})$$

where $a_n, \bar{b}_n$ are elements of a Grassman algebra, the coefficients in the expansion of the classical fields $\psi(x), \bar{\psi}(x)$:

$$\psi(x) = \sum a_n \varphi_n(x), \quad \bar{\psi}(x) = \sum \varphi_n^+(x) \bar{b}_n, \quad (\text{A.8})$$

in terms of the normalized eigenvectors of the hermitian operator $\not{D}$:

$$\begin{aligned} \not{D} \varphi_n(x) &= \lambda_n \varphi_n(x), \\ \langle \varphi_n | \varphi_m \rangle &= \int d^2x \varphi_n^+(x) \varphi_m(x) = \delta_{nm}. \end{aligned} \quad (\text{A.9})$$

The new fields $\chi(x), \bar{\chi}(x)$ defined in eq. (A.1) can also be expanded in terms of the φ_n 's:

$$\chi(x) = \sum a'_m \varphi_m(x), \quad (\text{A.10})$$

where

$$a'_m = \sum c_{mn} a_n, \quad (\text{A.11})$$

with

$$c_{mn} = \langle \varphi_m | U_5^{-1} | \varphi_n \rangle, \quad (\text{A.12})$$

and the jacobian factor is given by the relation

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = \det^{-2}(\mathcal{C}_{mn}) \mathcal{D}\bar{\chi}\mathcal{D}\chi. \quad (\text{A.13})$$

In order to compute $\det \mathcal{C}_{mn}$ we shall consider a one-parameter dependent γ_5 transformation:

$$\psi(x) = U_5(\alpha)\chi(x), \quad \bar{\psi}(x) = \bar{\chi}(x)U_5(\alpha), \quad (\text{A.14})$$

where

$$U_5(\alpha) = e^{\alpha\gamma_5\phi(x)}, \quad (\text{A.15})$$

and α is a real parameter to be varied from 0 to 1, allowing us to grow the whole transformation (A.1) by iteration of the infinitesimal one:

$$1 + \gamma_5\phi(x) \delta\alpha.$$

Now, at each stage of the transformation one has to consider the ‘‘covariant derivative’’:

$$\mathcal{D}(\alpha) = U_5(\alpha)\mathcal{D}U_5(\alpha) = i\hat{\partial} + g\mathcal{A}(\alpha), \quad (\text{A.16})$$

with

$$\mathcal{A}(\alpha) = \mathcal{A} + \frac{\sqrt{N}}{g} U_5(\alpha)\mathcal{D}U_5(\alpha). \quad (\text{A.17})$$

Of course, for $\alpha = 0$ and $\alpha = 1$, $\mathcal{D}(\alpha)$ satisfies:

$$\mathcal{D}(0) = \mathcal{D}, \quad \mathcal{D}(1) = \mathcal{D}'. \quad (\text{A.18})$$

The eigenvectors to be considered at each stage are then

$$\mathcal{D}(\alpha)\varphi_n(x; \alpha) = \lambda_n\varphi_n(x; \alpha). \quad (\text{A.19})$$

In order to compute (A.12) let us consider the matrix

$$B_{pn}(\beta, \alpha) = \langle \varphi_p(x; \beta + \alpha) | U_5^{-1}(\alpha) | \varphi_n(x; \alpha) \rangle, \quad (\text{A.20})$$

which satisfies

$$\frac{d}{d\alpha} \log \det B(0; \alpha) = \frac{\log \det B(\alpha; \delta\alpha)}{\delta\alpha}. \quad (\text{A.21})$$

Using the relation

$$|\varphi_n(x; \alpha + \delta\alpha)\rangle \approx |\varphi_n(x; \alpha)\rangle + \sum_{p \neq n} b_{np} |\varphi_p(x; \alpha)\rangle \delta\alpha, \quad (\text{A.22})$$

then

$$\frac{d}{d\alpha} \langle \log \det B(0; \alpha) \rangle = -w(\alpha), \quad (\text{A.23})$$

where

$$w(\alpha) = \sum_n \langle \varphi_n(x; \alpha) | \gamma_5 \phi(x) | \varphi_n(x; \alpha) \rangle, \quad (\text{A.24})$$

and hence by integration over α of (A.23) we get

$$\det \langle \varphi_p(x; 1) | U_5^{-1} | \varphi_p(x; 0) \rangle = \exp \left[\int_0^1 w(\alpha) d\alpha \right].$$

Now, since $\det \langle \varphi_p(x; 1) | \varphi_p(x; 0) \rangle = 1$,

$$\det C_{mn} = \det \langle \varphi_p(x; 1) | U_5^{-1} | \varphi_p(x; 0) \rangle = \exp \left[\int_0^1 w(\alpha) d\alpha \right]. \quad (\text{A.25})$$

In order to compute $w(\alpha)$ we write (A.24) in the form:

$$w(\alpha) = \int d^2x \operatorname{tr}^c \left[\phi(x) \sum_n \varphi_n^+(x; \alpha) \gamma_5 \varphi_n(x; \alpha) \right], \quad (\text{A.26})$$

and regularize the ill-defined sum [17]. The result is

$$w(\alpha) = \frac{g}{4\pi\sqrt{N}} \epsilon_{\mu\nu} \int d^2x \operatorname{tr}^c \phi(x) \left\{ \partial_\mu A_\nu(\alpha) - \partial_\nu A_\mu(\alpha) - \frac{ig}{\sqrt{N}} [A_\mu(\alpha), A_\nu(\alpha)] \right\}. \quad (\text{A.27})$$

Thus, the exponent W appearing in the jacobian,

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = \exp(-2W) \mathcal{D}\bar{\chi}\mathcal{D}\chi, \quad (\text{A.28})$$

$$W = \int_0^1 w(\alpha) d\alpha, \quad (\text{A.29})$$

takes the form

$$W = \frac{g}{2\pi} \int d^2x \int_0^1 d\alpha \epsilon_{\mu\nu} \phi(x) \partial_\mu \left[A_\nu + \frac{\alpha}{g} \epsilon_{\nu\rho} \partial_\rho \phi \right], \quad (\text{A.30})$$

for the abelian case [12]. For abelian vector fields decomposed in transverse and longitudinal parts,

$$A_\nu = -\frac{1}{g} \epsilon_{\nu\rho} \partial_\rho \phi + \frac{1}{g} \partial_\nu \eta, \quad (\text{A.31})$$

one can easily integrate over α , obtaining

$$J_F = \exp \left\{ -\frac{1}{2\pi} \int (\phi \partial_\mu \partial^\mu \phi + \epsilon_{\mu\nu} \phi \partial^\mu \partial^\nu \eta) d^2x \right\}. \quad (\text{A.32})$$

For the SU(2) case, making a decomposition of the kind (A.31) except that the transverse part is replaced by the “decoupling” one:

$$A^d = \gamma_\mu A_\mu^d \cdot \mathbf{t} = \frac{i\sqrt{N}}{g} (\partial U_5) U_5^{-1} \quad (\text{A.33})$$

the jacobian is given by [13]

$$J_F = \exp \left\{ -\frac{\epsilon_{\mu\nu}}{4\pi} \int d^2x \phi \cdot \left(\partial_\mu A_\nu - \partial_\nu A_\mu + A_\mu \wedge A_\nu \left(\frac{1}{\phi \tanh \phi} - \frac{1}{\sinh^2 \phi} \right) \right) \right\}. \quad (\text{A.34})$$

For a general A_μ of the form

$$A_\mu = U^{-1}(\eta) A_\mu^d U(\eta) + \frac{i\sqrt{N}}{g} U^{-1}(\eta) \partial_\mu U(\eta), \quad (\text{A.35})$$

eq. (A.27) is much more complicated even in the SU(2) case.

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